Adaptive Trial Design for the RAS-Inhibitor Era: Combinations, Resistance, and Biomarker-Guided Therapy

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Today’s argument

> **New RAS therapies in PDAC need new kinds of trials.**

We know how to **optimize combination doses**.

We know how to **adapt trials to emerging resistance**.

> **The challenge is making these designs practical enough to launch.**

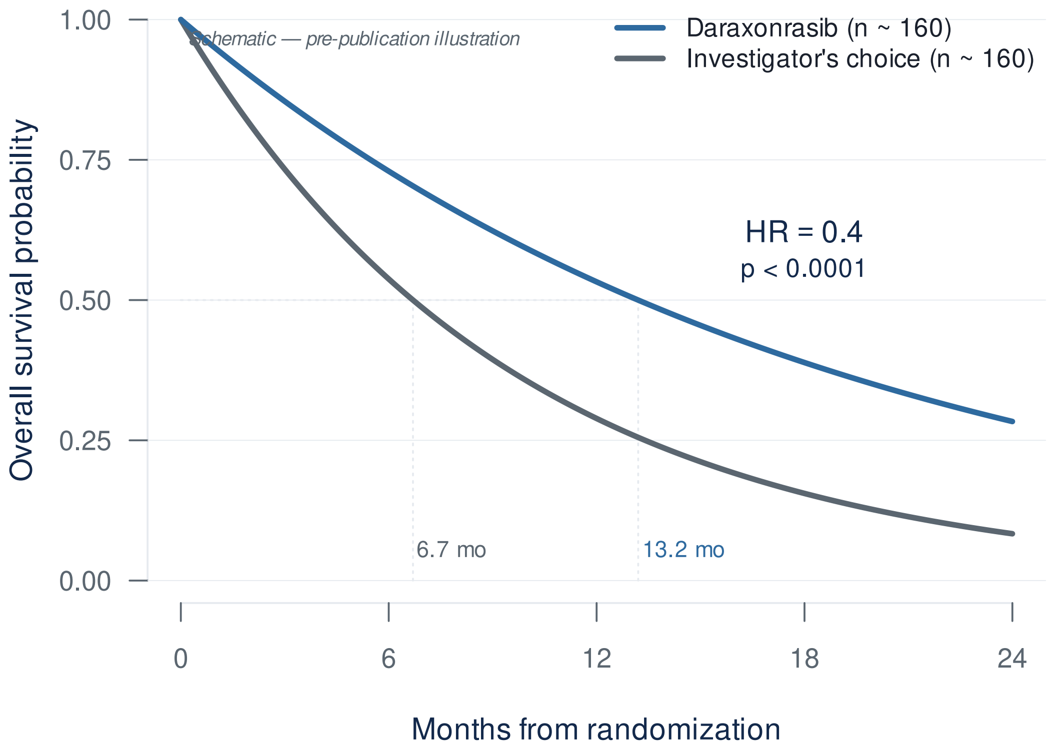
The clinical case

Why combinations, and why now?

The RAS revolution has arrived for PDAC

A major survival signal from direct RAS inhibition in PDAC.

- **RASolute 302** (2L mPDAC, *NEJM* 2026)[1]:
  - **Median OS: 13.2 vs 6.7 months**
  - **HR 0.40 — 60% reduction in risk of death** ($p < 0.0001$)
- Daraxonrasib (RMC-6236) — oral, multi-selective RAS(ON) inhibitor
- ASCO 2026 Plenary, May 31 (Wolpin)



…but monotherapy has visible ceilings

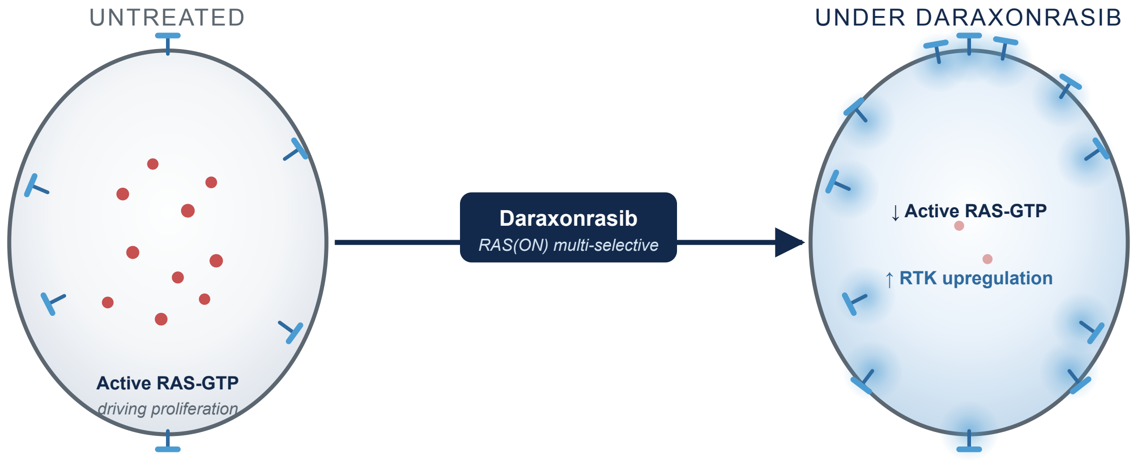
RMC-6236-001 (PDAC cohort, NEJM 2026)[2]

48% of monotherapy patients require dose modification at 300 mg — a strong signal of **limited tolerability margin**. Combinations can layer **overlapping, non-additive** toxicity that single-agent data don't reliably predict — which is why the combination needs its own dose-finding.

Finding the right combination dose becomes the next challenge.

Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK upregulation** on PDAC cell surface — the mechanistic case for biomarker-driven, resistance-preemptive combinations[3]
- **1L combination cohort** (daraxonrasib **200 mg** + GnP, n=40)[4]:
  - ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**



RTK upregulation is simultaneously a resistance mechanism and a rational combination target

> *The same adaptation that creates resistance may create a therapeutic vulnerability.*

The dose-finding problem

Why MTD only may not be best for RAS combinations, and what replaces it.

Why combinations break the rules you grew up on

- **Rivière's review of 162 combination Phase I trials: 88% used 3+3** even when both drugs were escalated[5,6].
- **Why 3+3 doesn't fit combinations.** Built for single-agent cytotoxics[7], it assumes monotonic dose-toxicity, Cycle-1-observable DLTs, and a single dose dimension — combinations strain all three.
- **The two-drug dose map is never built.** In practice: pick A's dose, fix it, add B. Two monotherapy escalations run in series; the joint efficacy/toxicity behavior across (A, B) pairs (the "joint surface") is never characterized.

Two-drug dose map: how efficacy and toxicity change across every (A, B) dose pair — not just along one drug's axis.

The Amgen case: sotorasib + pembrolizumab

A real-world example of why combination toxicity can surprise us.

Same sponsor, two different designs:

- **Monotherapy (CodeBreaK 100):** Bayesian model-based, single-agent surface
- **Combination:** IO dose fixed, only sotorasib varied — joint surface never modeled

The defining toxicity:

- Severe **immune-mediated** hepatotoxicity, not predictable from the single agent
- **88%** of Grade 3-4 events outside the Cycle 1 DLT window[8]

> **What MTD-only dose-finding was never built to handle, all at once:** joint surface unsearched, non-additive toxicity, late-onset toxicity.

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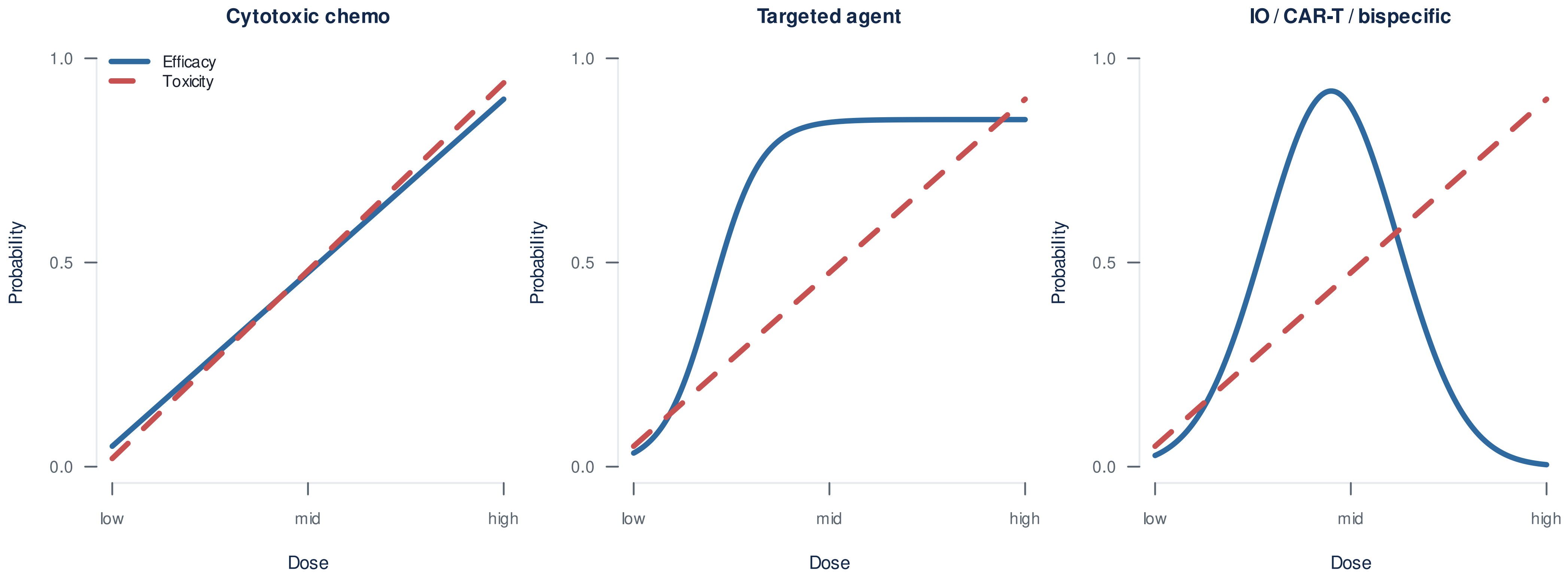
Why MTD-only dose-finding doesn’t fit targeted agents and immunotherapies

CYTOTOXIC | **TARGETED** | **IMMUNE**



- **Pembrolizumab** — efficacy plateaus at the lowest dose tested; MTD never reached[9]
- **CAR-T** — efficacy *rises then falls* (umbrella); high doses drive severe CRS/ICANS, not benefit
- **A RAS combination spans both shapes.** Neither rewards titrating to toxicity.

The cost of MTD-only dose-finding

- **Sotorasib** (KRAS G12C, first-in-class, approved 2021)[10]:
  - 960 mg selected as the **highest dose tested**, no DLTs — a more-is-better default, not an MTD
- **MTD ≠ optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions[11]

240 mg ≈ 960 mg on PFS

*FDA-required post-marketing comparison: similar PFS at **one-quarter** the exposure[12].*

Same drug class as daraxonrasib.

Project Optimus: the regulatory ground has shifted

How OBD selection actually works

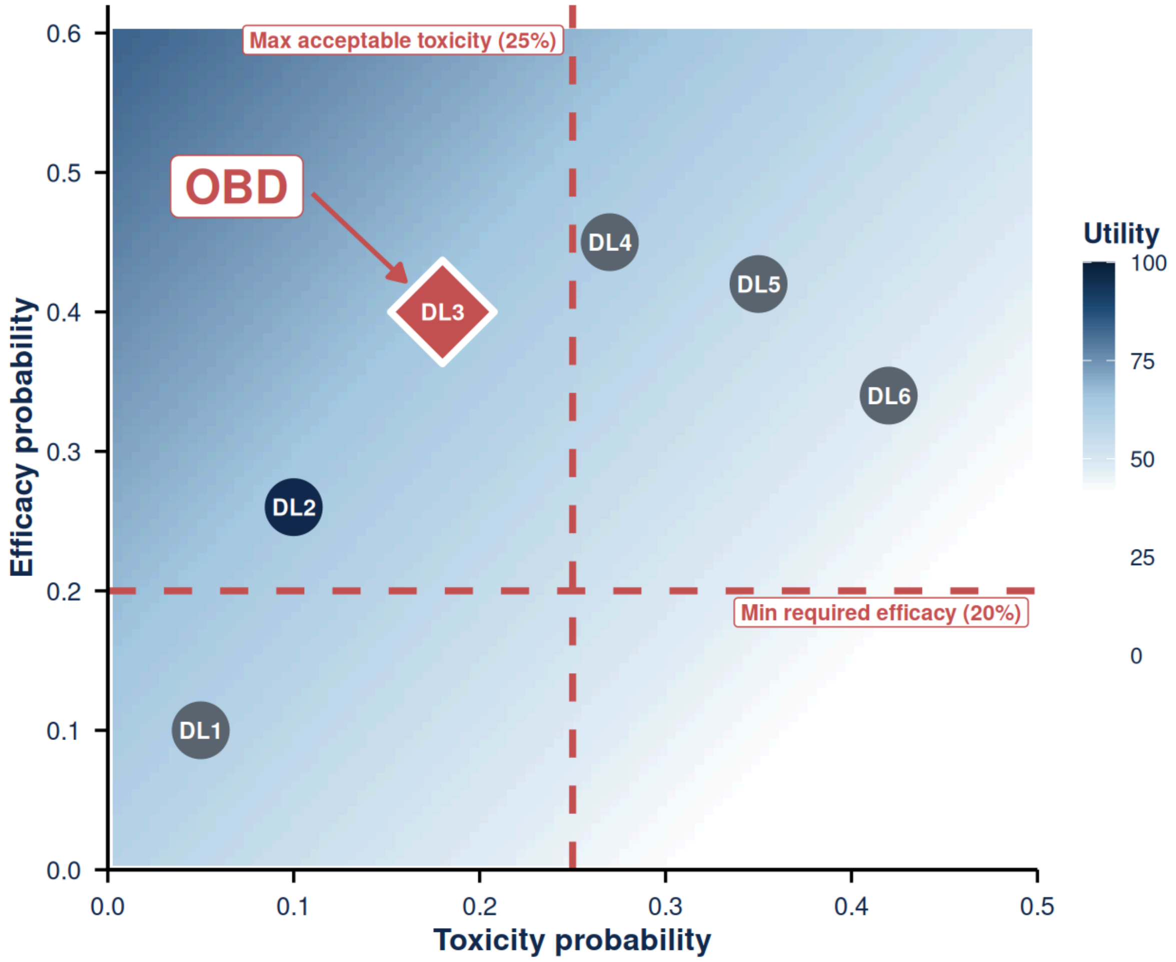
Instead of escalating until toxicity, choose the dose with the best benefit-risk profile.

“How much efficacy would you trade for additional toxicity?”

The team writes that utility table down up front. The design then searches inside the safe envelope for the dose that **maximizes benefit-risk**, not toxicity.

Elicited utility table (Zhou 2019 framework)[14]:

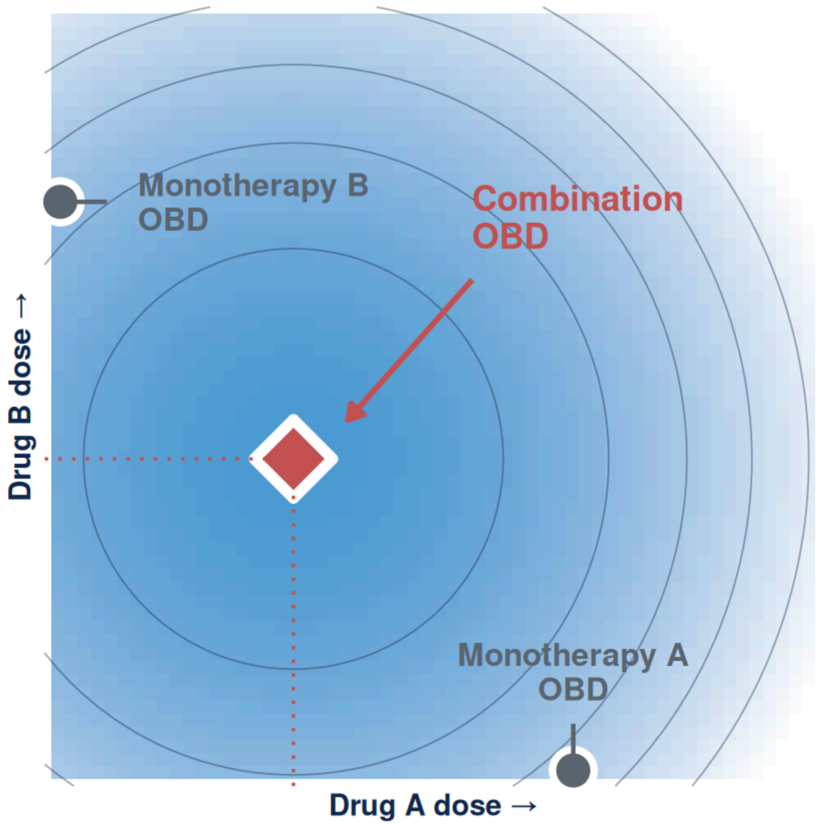
	TOX	NO TOX
Response	40	**100**
No response	**0**	60



***OBD** = admissible dose (inside red lines) with the highest utility.*

Combination dose selection in RAS is harder

Combinations make MTD-only dose-finding worse on every axis.



Conceptual schematic of the benefit-risk surface. The combination OBD sits below both single-agent OBDs; single-agent data alone cannot reconstruct the contour.

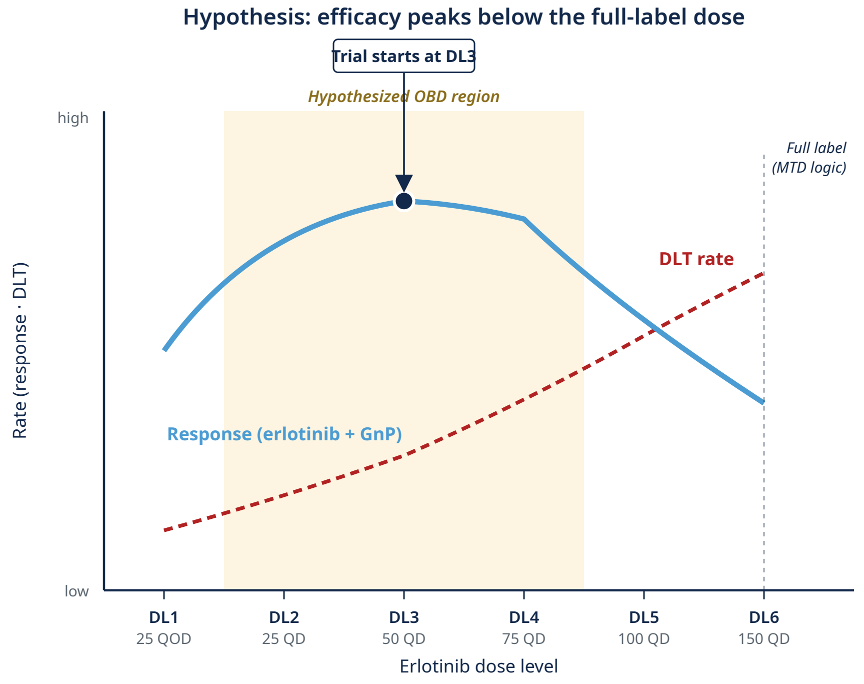
- The combination OBD can sit **below both single-agent OBDs**
- Single-agent data **cannot reconstruct** the joint surface
- For RAS, the **toxicity margin is already spent** at the monotherapy dose

> **Fix what you know. Search what you don't.**

RMC-GI-102 (daraxonrasib + GnP) fixed daraxonrasib at 200 mg without searching the joint surface[4].

Finding the right erlotinib dose in basal-like PDAC

- **Clinical question.** Could *lower*-dose erlotinib work *better* than full-dose erlotinib in basal-like PDAC, when combined with GnP?
- **Design.** First find the safe dose range, then choose the dose with the best response/toxicity trade-off[14,17].
- **Why it matters.** This is an OBD-below-MTD question; 3+3 cannot answer it.

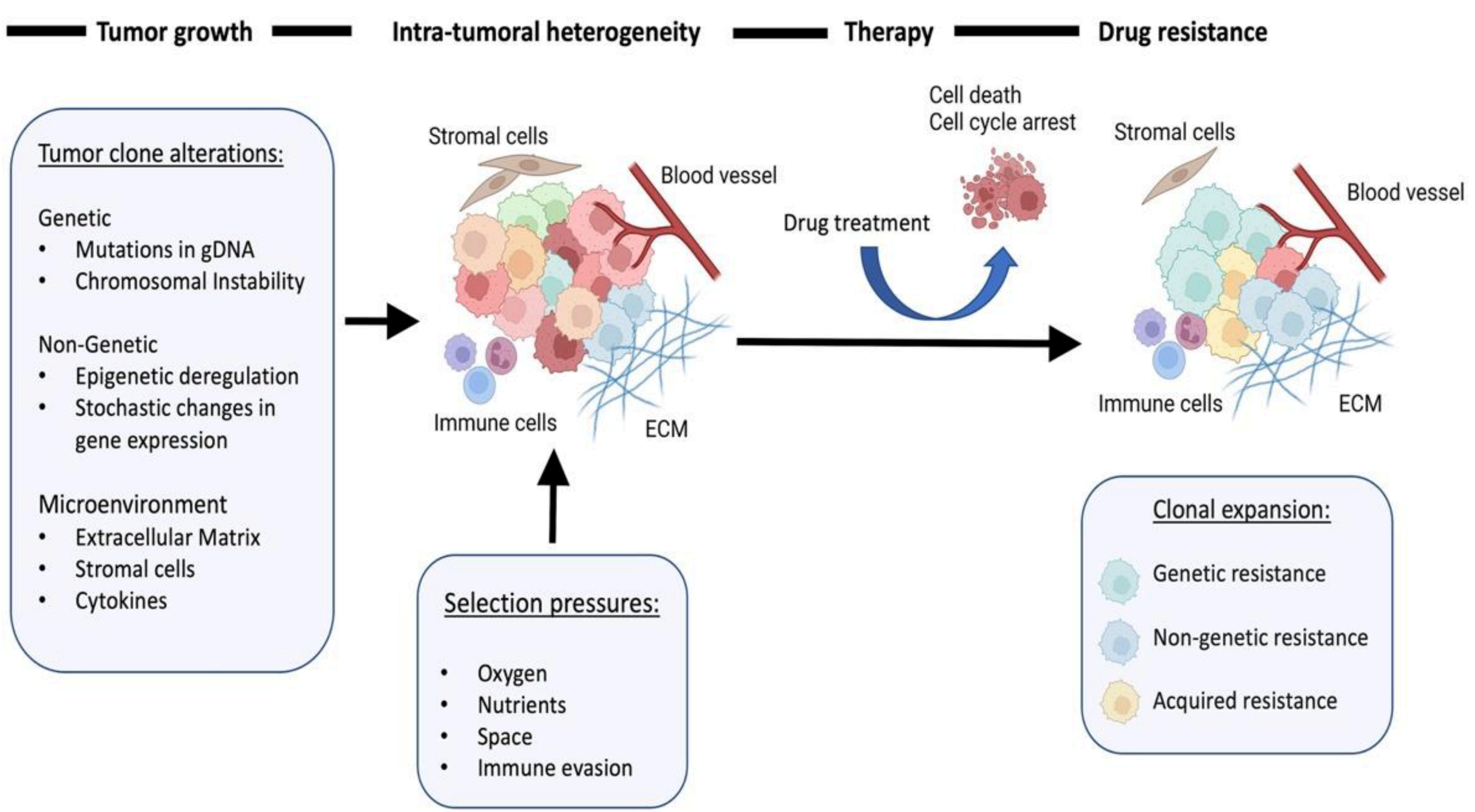


Trial: LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by PurIST[18] · combination erlotinib + biweekly GnP · OBD-below-MTD hypothesized from prior data[19] · **up to 52 patients; ~28 expected at the OBD.**

Resistance is the next problem

Phase 2 onward: *each patient is a moving target*

Resistance is a dynamic process

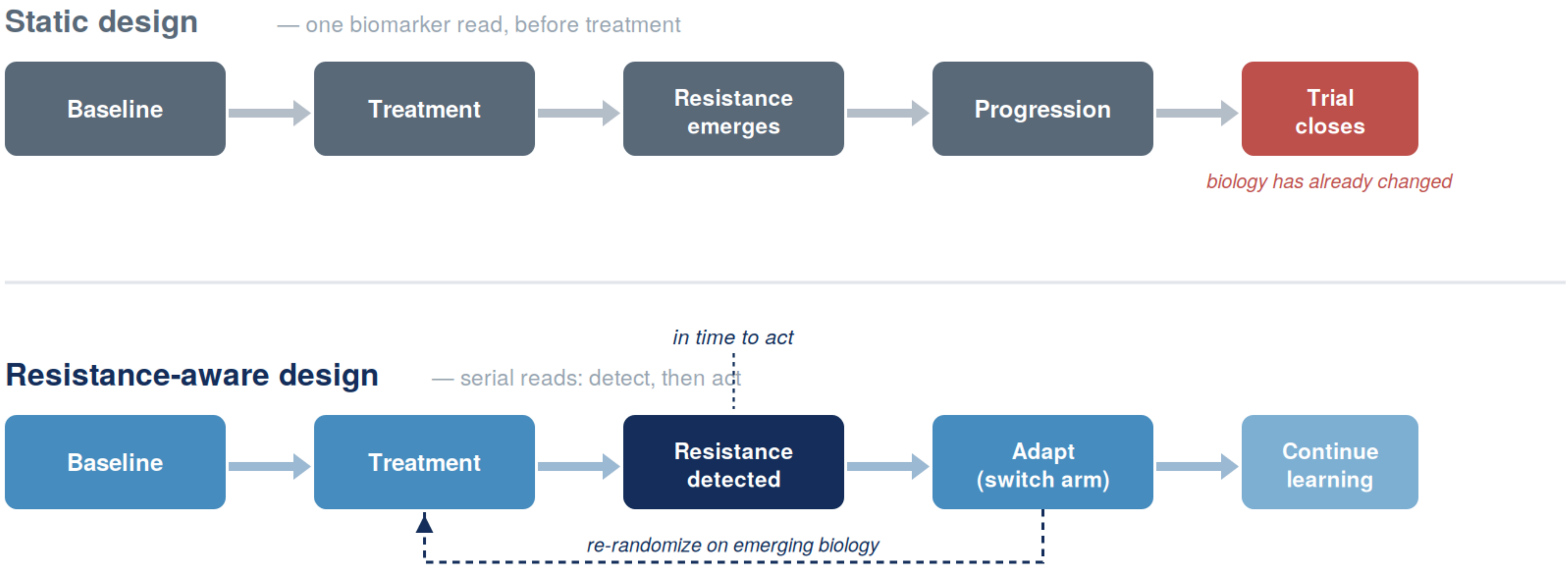


Adapted from Karagiannis & Rampias, Cancers 2022[20].

Resistance mechanisms emerge under RAS-inhibitor pressure

> **Daraxonrasib monotherapy in mPDAC** (n=44, paired ctDNA): **59% show RAS-pathway resistance at progression — KRAS amplification in 36%**[21]

Why classical trial designs miss resistance



A static read measures a moving target.

Longitudinal detection + biomarker-guided arm-swaps keep the trial in step with the tumor.

Detecting resistance takes two reads, not one

Different resistance mechanisms become visible through different assays.

RESISTANCE TYPE	WHAT YOU NEED TO MEASURE IT
Genomic KRAS amplification, NRAS / BRAF / MAP2K1 alterations	**ctDNA** *serial liquid biopsy, minimally invasive*
Cell-state / plasticity basal-to-classical switching, partial EMT	**biopsy + transcriptomics + imaging**

A resistance-aware trial has to run both, during enrollment, not at progression.

Genomic and cell-state resistance are invisible to each other's assay.

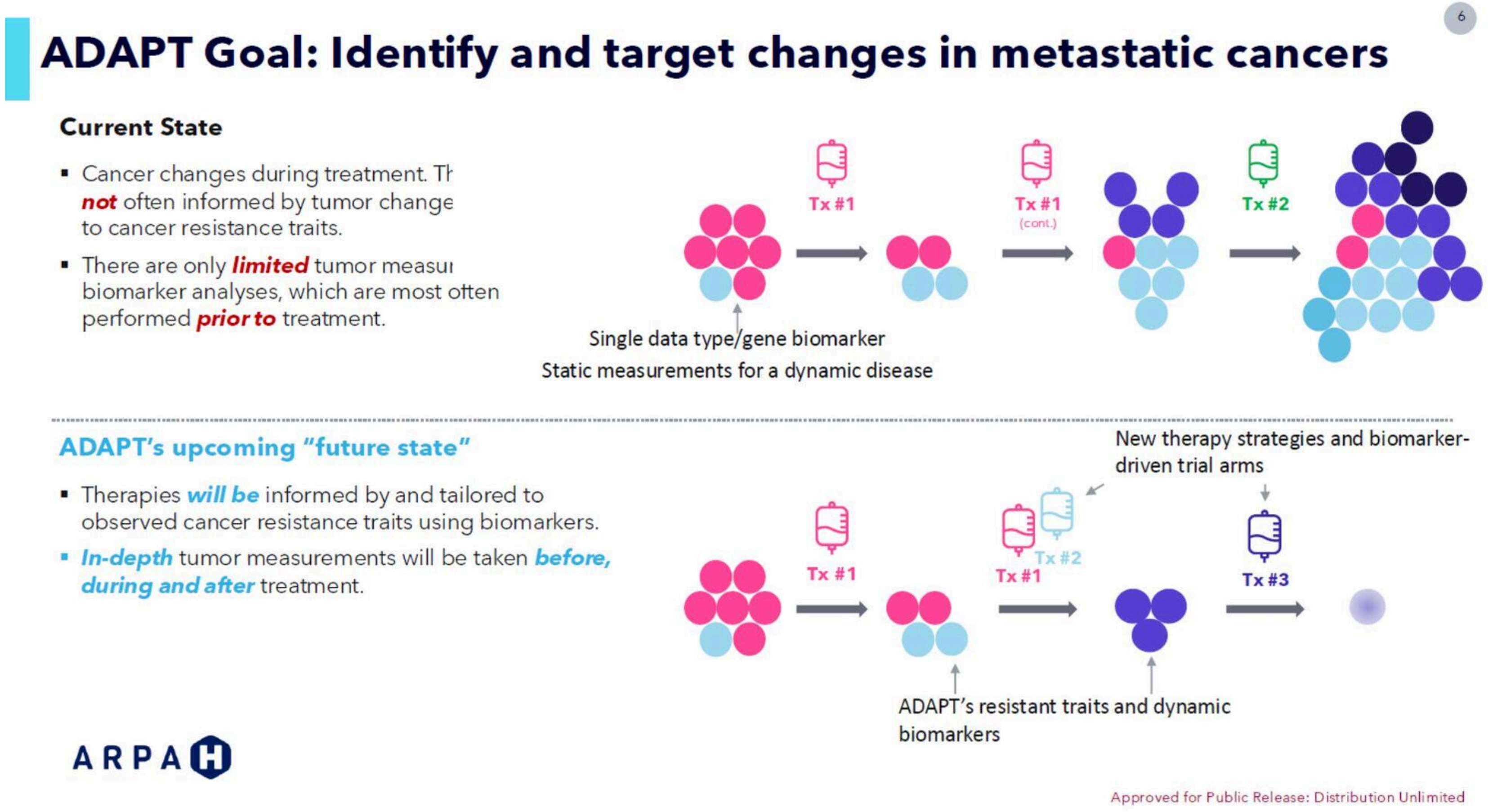
ADAPT / EVOLVE: what a resistance-aware trial looks like

ARPA-H's resistance-tracking template.

ARPA-H's flagship adaptive resistance-monitoring platform.

- Longitudinal resistance detection
- Biomarker-guided adaptation
- Adaptive arm-swaps on emerging biology

*ARPA-H **ADAPT** → **EVOLVE** (UNC-led; Carey MPI, Rashid MPI).* ***Enrolling.***

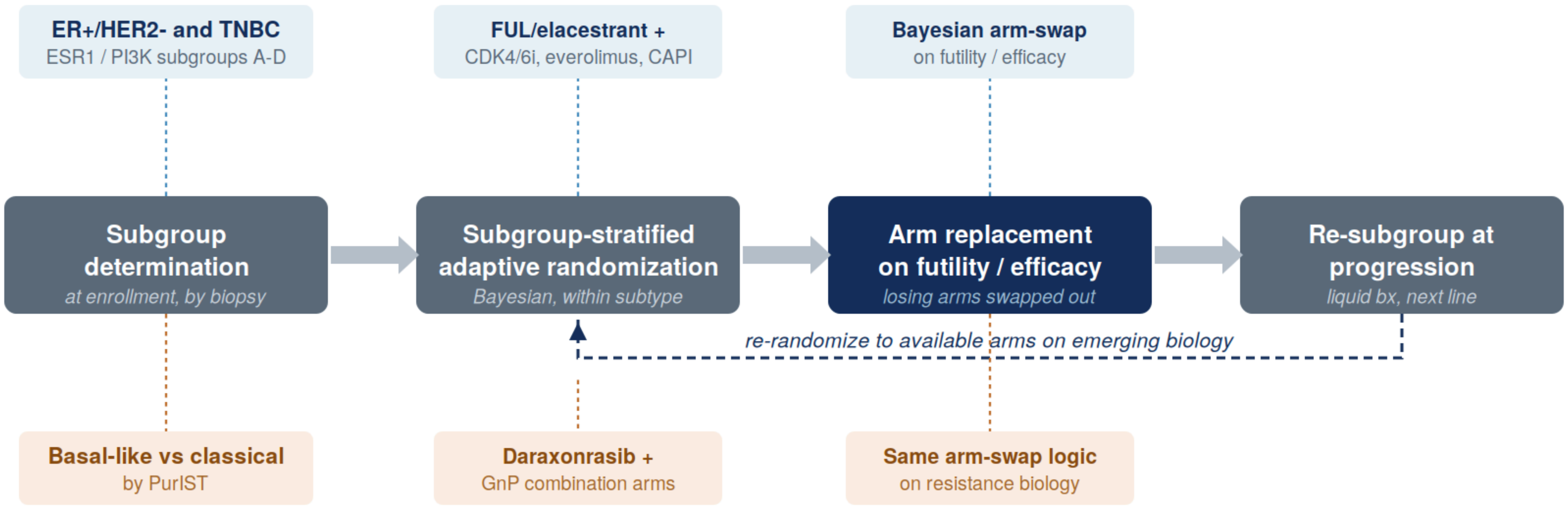


ARPA-H ADAPT — current state (top) vs dynamic-biomarker state (bottom).

Same architecture, different disease

The EVOLVE-BDT flow is reusable: only the biomarker and the drugs change.

EVOLVE-BDT — breast (real, enrolling) **PDAC analog (PurIST-defined, hypothetical)**



The same adaptive framework applies to PurIST-defined PDAC subtypes and evolving resistance biology.

Swap the biomarker and the drugs; the calibration machinery and decision rules carry over.

What it would take to actually run these trials

Four things any such trial has to do at once.

Stop ineffective therapies earlier

Bayesian futility / efficacy boundaries at each interim

Learn who benefits while the trial is running

Pre-specified signatures + agnostic discovery window

Track resistance as it emerges

Serial ctDNA, imaging, biopsies on every patient

Test several biomarker groups under one protocol

Biomarker-guided cohorts within the same protocol

Everything I've just described already exists statistically. So why don't we see these trials everywhere?

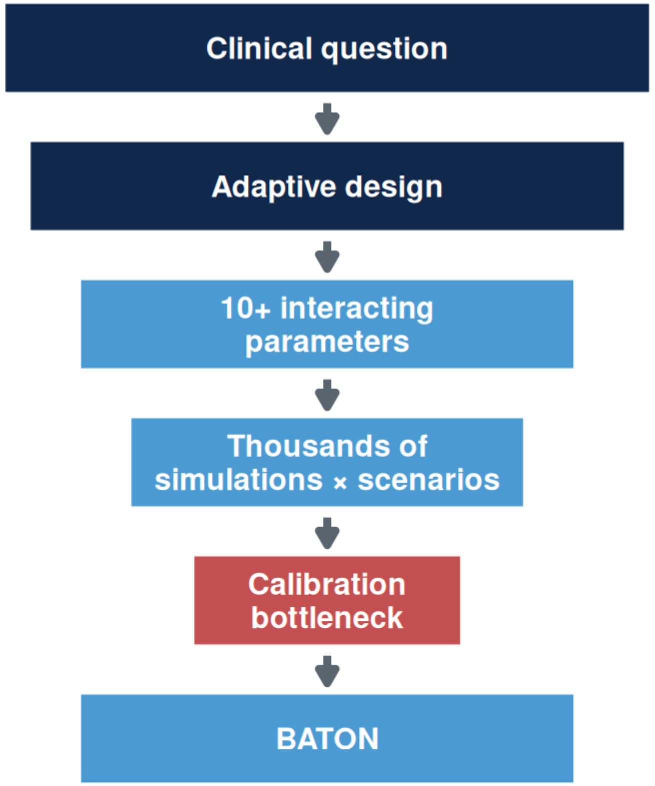
Putting these into practice

Why don’t we see more of these trials?

Funding, accrual, and biopsy logistics are real.

But the barrier that quietly kills good designs — and the one we can actually remove — is calibration.

Before an adaptive trial can open, **thousands of simulations** are needed to prove it is safe, efficient, and scientifically reliable.
That process — not the design itself — is often the bottleneck.



Methods adopted. Dose-optimization plans — where calibration cost bites — not[22].

Our group: several months per design. For combined dose + resistance designs, hand-tuning is infeasible.

Calibration has to hit four goals at once

Designing an adaptive trial requires achieving all four simultaneously.

1. FDA compliance Power ≥ 0.80 Type I ≤ 0.10	**2. Minimize patients** Smaller E[N] Efficient use of scarce patients
3. Stop bad early Futility stopping Protect patients from inactive arms	**4. Accelerate good** Efficacy stopping Faster access when drug works

Of thousands of plausible designs, only a small fraction satisfies all four — and you cannot find it by hand.

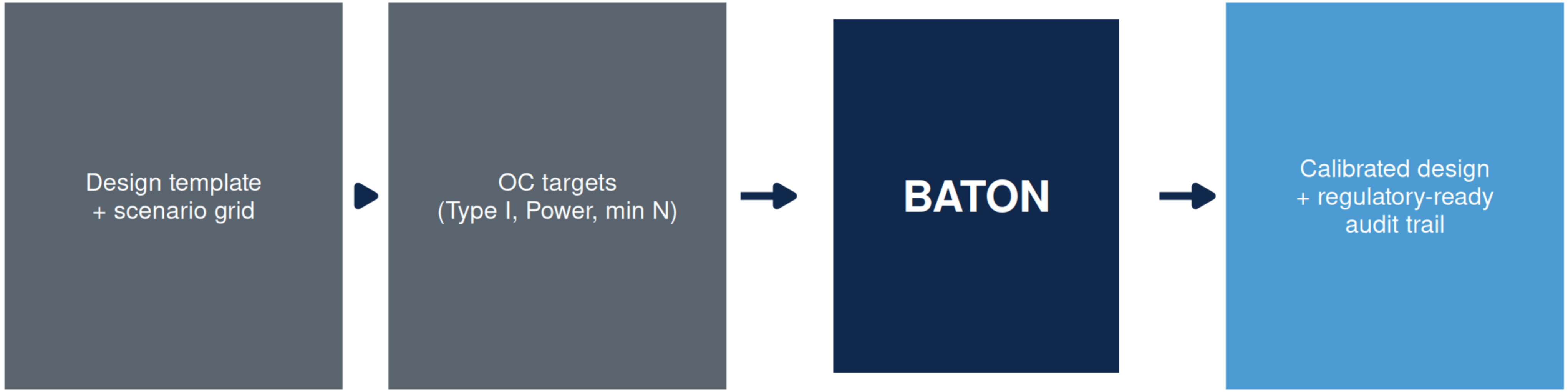
"We knew what trial to run.

We needed a way to make it launchable."

BATON: a way to make adaptive trials launchable

BATON automatically searches thousands of candidate designs and identifies those that meet FDA, efficiency, and scientific goals.

BATON = **B**ayesian **A**daptive **T**rial **O**ptimizatio**N**[23]



A BATON-calibrated design hits all four goals — for PANGEA Phase II

1. FDA compliance ✓

Type I 7.0% (target ≤10%)
Power 82.2% (target ≥80%)

2. Minimize patients ✓

~23 patients spared
(avg N 53 vs Max N 76)

3. Stop bad early ✓

78% early futility stop
when drug is inactive

4. Accelerate good ✓

81% early efficacy stop
when drug works

Far fewer evaluations than grid search — finds feasible designs grid search often misses entirely.

For this 4-parameter design: manual calibration takes days · BATON takes minutes — with a regulatory-ready audit trail.

What this means for a PDAC patient

Mr. Hernandez, 62, mPDAC, basal-like by PurIST, progressed on 1L GnP.

Traditional fixed Phase II

- All **76 patients** enrolled before any look
- Drug effect discovered only at study end
- **~36+ months** to read out

PANGEA Phase II (BATON-calibrated)

- **78% chance** of early futility stop **when the drug is inactive**
- Average patients enrolled: **~53** if inactive / **~58** if active
- **27.3 months** expected duration

Sooner answers. Fewer patients exposed to ineffective therapy. No sacrifice in statistical rigor.

PANGEA Phase II: calibrated by BATON

BATON identified a design that preserves power, controls false positives, and allows early stopping when erlotinib isn't helping.

- After Phase I OBD → **Phase II Bayesian adaptive RCT:** GnP + erlotinib (at OBD) vs GnP alone, primary PFS
- **Design assumes** reference 5.5 mo → 9.2 mo (HR 0.58)[24]
- **BATON-calibrated** four parameters under Type I ≤ 0.10, Power ≥ 0.80[23]

Achieved operating characteristics:

METRIC	VALUE
Power	**82.2%**
Type I error	7.0%
Max N	**76** (82 accrual w/ buffer)
Expected N under H_0	53.0
Expected N under H_1	58.1
Pr(early futility stop | H_0)	**77.9%**
Pr(early efficacy stop | H_1)	**81.4%**
Expected duration	27.3 mo

H_0 = drug inactive · H_1 = drug active (HR 0.58). Read as: 78% early-stop when inactive; 81% early-stop when it works.

Why max N = 76, not lower: *caps set too low silently suppress early-efficacy stopping. BATON surfaced the trade-off curve; we chose 76 to preserve it.*

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What becomes possible

BATON calibrates **aggressive**, **conservative**, or **balanced** designs — and ports across GI applications:

PDAC
RAS-backbone combinations
OBD selection

CRC
Joint-surface novel-novel doublets

Basket / umbrella
Biomarker-stratified subtrials
shared decision rules

Robust calibration
Designs robust to uncertain trial assumptions

BATON is the infrastructure that makes adaptive trial design routine across the GI portfolio.

Adaptive, resistance-aware trials as the routine standard across GI — not the exception.

Take-home

1. **The RAS era needs new kinds of trials** — biomarker-guided, dose-optimized, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs can handle dose optimization, biomarker guidance, and resistance-aware adaptation.
3. **BATON removes one of the major barriers preventing these designs from reaching patients.**

Acknowledgments

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Questions